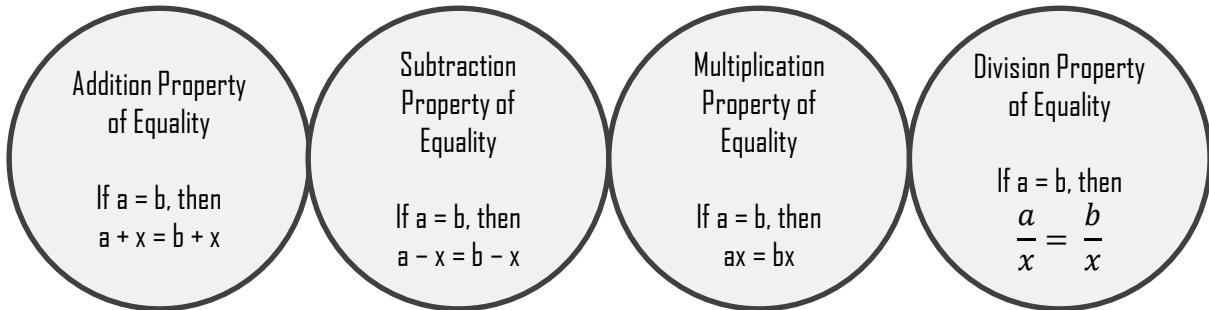
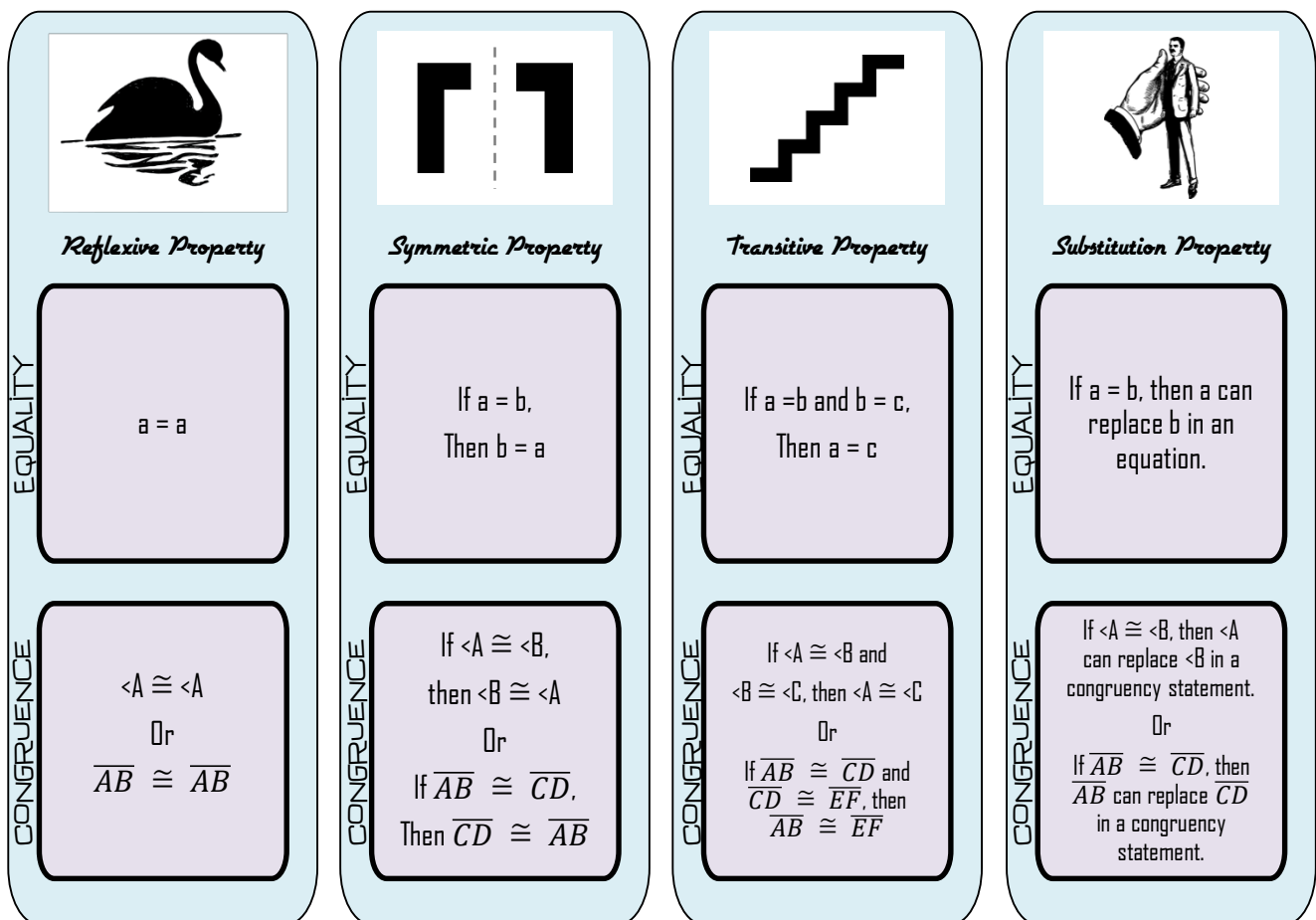


GRAPHIC ORGANIZER: PROPERTIES OF EQUALITY AND CONGRUENCE

Name: _____
Date: _____ Class: _____



BASIC PROPERTIES OF EQUALITY (For all nonzero real numbers)



PROPERTIES OF EQUALITY AND CONGRUENCE



PROPERTIES OF EQUALITY AND CONGRUENCE

Name: _____
Date: _____ Class: _____

Complete the diagrams. The boxes should be filled in with equations and congruency statements. The arrows should be filled in with the properties needed to get from one step to another.

1

$$14m + 12 = 44 - 2m$$

Addition Property of Equality

$$14m + 2m + 12 = 44 - 2m + 2m$$

$$\text{or } 16m + 12 = 44$$

$$\frac{16m}{16} = \frac{32}{16} \text{ or } m = 2$$

2

$$\angle TUV \cong \angle VWX,$$

$$\angle TUV \cong \angle PQR$$

Transitive Property of Congruence

Symmetric Property of Congruence

3

$$m\angle 2 = m\angle 3,$$

$$m\angle 3 = m\angle 4$$

$$m\angle 3 + m\angle 5 = m\angle 4 + m\angle 5$$

$$m\angle 2 + m\angle 5 = m\angle 4 + m\angle 5$$

Subtraction Property of Equality

In diagram 3, which property could get from the first step to the last more quickly?



PROPERTIES OF EQUALITY AND CONGRUENCE

Name: _____
Date: _____ Class: _____

Remember: In order to use the Transitive Property, the equivalent parts must each be an ENTIRE SIDE of the equation.



4

$$m\angle 1 + m\angle 2 = m\angle 5, \\ m\angle 2 = m\angle 3$$

$$m\angle 1 + m\angle 3 = m\angle 5$$

Transitive Property of Equality

$$m\angle 2 = m\angle 3$$

$$m\angle 2 + m\angle 7 = m\angle 3 + m\angle 7$$

Symmetric Property of Equality

Note: In a real proof, it would not really be very helpful to end up back where we started, but we are just practicing using the properties to justify each step.



5

$$\angle DGF \cong \angle FHK, \\ \angle FEC \cong \angle ABD, \\ \angle FHK \cong \angle FEC$$

$$\angle DGF \cong \angle ABD$$

Note: You may always go back to the beginning to find and use more of the given statements at any point



PROPERTIES OF EQUALITY AND CONGRUENCE WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Identify the property.

1. Given that $m\angle BCA = m\angle DBC$ and $m\angle BCA = m\angle ADB$, we can conclude that $m\angle DBC = m\angle ADB$.
2. Given that $\overline{PQ} \cong \overline{TU}$, we can conclude that $\overline{TU} \cong \overline{PQ}$.
3. $\triangle ABC \cong \triangle ABC$
4. Given that $x = 3y$, we can conclude that $x - 13 = 3y - 13$.
5. Given that $m\angle 2 = m\angle 6$, we can conclude that $m\angle 2 + m\angle 4 = m\angle 6 + m\angle 4$.
6. Given that $\angle MNO \cong \angle LPR$, we can rewrite the statement $\angle MNO \cong \angle DEF$ as $\angle LPR \cong \angle DEF$.
7. Given that $c = a$, we can conclude that $7c = 7a$
8. Given that $\overline{GH} \cong \overline{VW}$ and $\overline{TU} \cong \overline{GH}$, we can conclude that $\overline{VW} \cong \overline{TU}$.
9. Given that $XY = MN$ and $XY + YZ = XZ$, we can state that $MN + YZ = XZ$.
10. If $4k = 18$ then $18 = 4k$.



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PROPERTIES OF EQUALITY AND CONGRUENCE WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

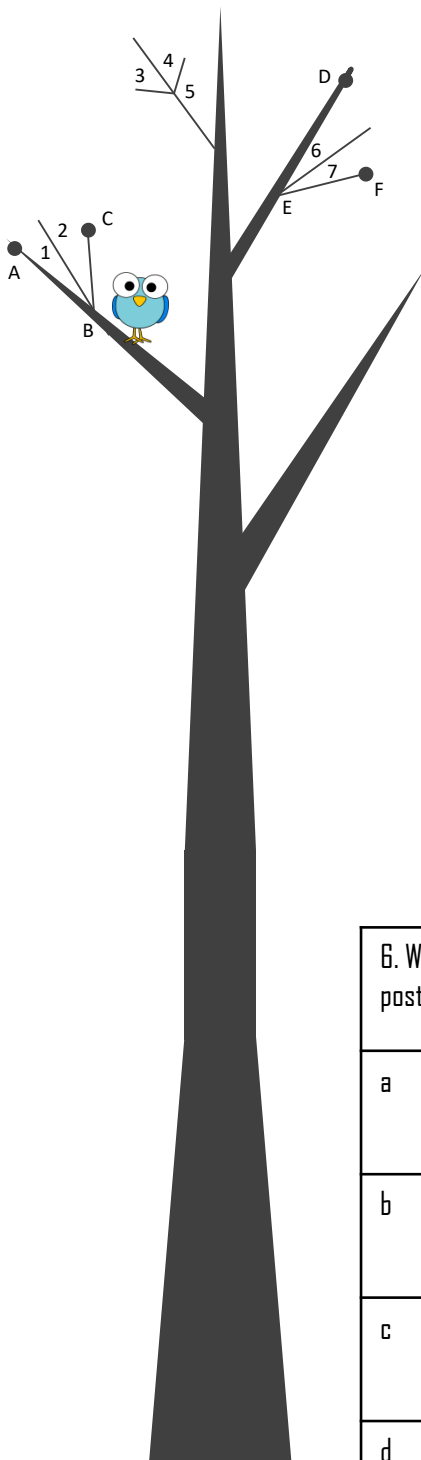
Explain the difference between using Substitution and the Transitive Property.
Give one example in which either method can be used to reach the same conclusion and one example in which the Substitution cannot be accomplished using the Transitive Property.



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SEGMENT ADDITION POSTULATE AND ANGLE ADDITION POSTULATE

Name: _____
Date: _____ Class: _____



- Given that the distance from the ground to the lowest branch is 39 meters and the distance from the ground to the branch with the bird is 46 meters, find the distance between the lowest branch and the branch with the bird.
- Given that the distance from the branch with the bird to the top of the tree is 15 meters, find the height of the tree.
- Given that the measure of angle 1 is 28 degrees and the measure of angle 2 is 55 degrees, find the measure of angle ABC.
- Given that the measure of angle DEF is 62 degrees and angle 6 is congruent to angle 7, find the measure of angle 6.
- Given that the measure of angle 3 is 23 degrees, the measure of angle 5 is 134 degrees, and the measure of the large angle made up of all three angles 3, 4, and 5 is 221 degrees, Find the measure of angle 4.

6. Write an equation that can be used to find the requested measure and identify which postulate you used.

a		Find $m\angle MOP$.	Equation: _____	Postulate: _____
b		Find XZ.	Equation: _____	Postulate: _____
c		Find $m\angle AED$.	Equation: _____	Postulate: _____
d		Find SW.	Equation: _____	Postulate: _____



TWO- COLUMN STRUCTURE FOR PROOFS: ALGEBRA PROOFS

Name: _____
Date: _____ Class: _____

Fill in the missing justifications in each proof.

1. Given: $a = 2b + 6$
 $a = 9b - 8$

Prove: $b = 2$

2. Given: $\frac{x}{2} = y + 5$
 $y = 3x$

Prove: $x = -2$



	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1	$a = 2b + 6$		1	$\frac{x}{2} = y + 5$	
2	$a = 9b - 8$		2	$y = 3x$	
3	$2b + 6 = 9b - 8$		3	$\frac{x}{2} = 3x + 5$	
4	$6 = 7b - 8$		4	$x = 6x + 10$	
5	$14 = 7b$		5	$-5x = 10$	
6	$2 = b$		6	$x = -2$	
7	$b = 2$		7		
8			8		

3. Given: $3w = t$
 $2r = s$
 $t = 2r$

Prove: $3w = s$

4. Given: $3b + d = f$
 $d = 2b$
 $f = g$

Prove: $g = 5b$

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1	$3w = t$		1	$3b + d = f$	
2	$2r = s$		2	$d = 2b$	
3	$t = 2r$		3	$f = g$	
4	$3w = 2r$		4	$3b + 2b = f$	
5	$3w = s$		5	$5b = f$	(simplified step 4)
6			6	$5b = g$	
7			7	$g = 5b$	
8			8		



TWO- COLUMN STRUCTURE FOR PROOFS: ALGEBRA PROOFS

Name: _____
Date: _____ Class: _____

Complete each proof.

5. Given: $5c + 1 = 2c - 8$

Prove: $c = -3$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

6. Given: $m = n + 5$
 $2m = n$

Prove: $m = -5$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

7. Given: $g = 2h$
 $g + h = k$
 $k = m$

Prove: $m = 3h$



	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

8. Given: $ab = 8a$
 $b + d = 20$

Prove: $d = 12$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		



ALGEBRA PROOFS

WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Given: $w + 4 = 7w - 2$
 $v = w$

Prove: $v = 1$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

ALGEBRA PROOFS

WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

Given: $c + d = f$
 $f = k$
 $c = d$

Prove: $k = 2c$

	STATEMENT	JUSTIFICATION
1		
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8		



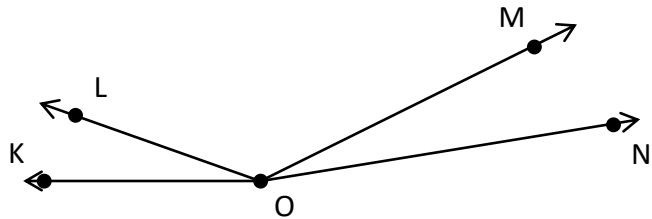
GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES

Name: _____
Date: _____ Class: _____

1

Given: $m\angle KOL = m\angle MON$

Prove: $m\angle KOM = m\angle LON$

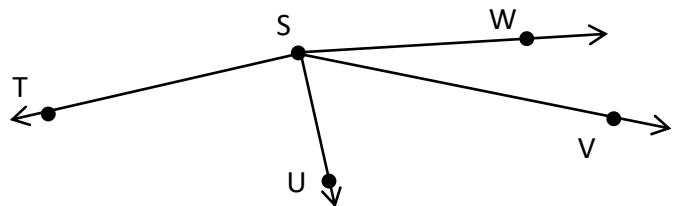


	STATEMENT	JUSTIFICATION
1		
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2

Given: $\overrightarrow{SU}$ bisects $\angle TSW$

Prove: $m\angle USV + m\angle VSW = m\angle TSU$



	STATEMENT	JUSTIFICATION
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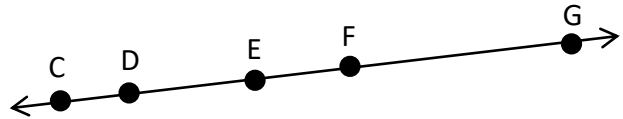
GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES

Name: _____
Date: _____ Class: _____

3

Given: $\overline{CD} \cong \overline{EF}$
 $\overline{DE} \cong \overline{FG}$

Prove: $\overline{CE} \cong \overline{EG}$

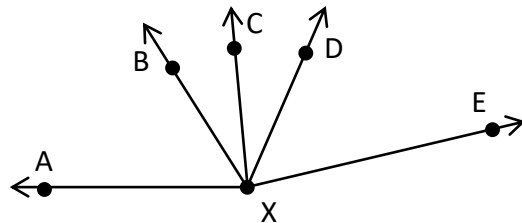


	STATEMENT	JUSTIFICATION
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4

Given: $\overrightarrow{XB}$ bisects $\angle AXE$
 $\overrightarrow{XD}$ bisects $\angle CXE$

Prove: $m\angle DXE + m\angle AXB = m\angle BXD$



	STATEMENT	JUSTIFICATION
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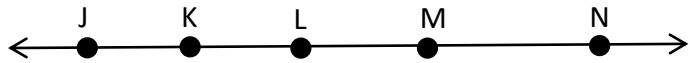


GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Given: $\overline{JM} \cong \overline{KN}$

Prove: $\overline{JK} \cong \overline{MN}$



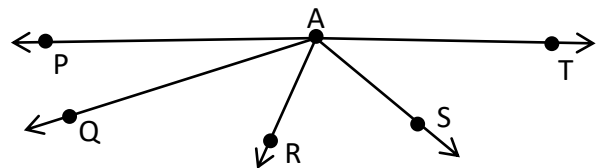
	STATEMENT	JUSTIFICATION
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GEOMETRY PROOFS USING DEFINITIONS AND POSTULATES WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

Given: $\angle PAQ \cong \angle RAS$, $\angle RAS \cong \angle SAT$

Prove: $\angle PAS \cong \angle QAT$



	STATEMENT	JUSTIFICATION
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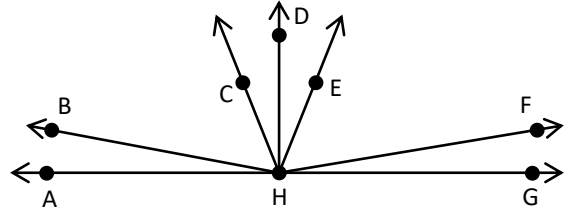


Name: _____

PROOF PUZZLE: CUT OUT STATEMENTS AND JUSTIFICATIONS AND SEE IF YOU CAN ARRANGE THEM TO DEVELOP A TWO COLUMN PROOF. YOU MIGHT NOT USE ALL OF THE PIECES. THERE MAY BE MORE THAN ONE CORRECT FINISHED PROOF.

Given: $m\angle AHB = m\angle FHG$, $m\angle CHD = m\angle DHE$, $m\angle BHD = m\angle DHF$

Prove: $m\angle CHF = m\angle BHE$



	STATEMENT	JUSTIFICATION
1		
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9		
10		
11		
12		
13		



STATEMENTS AND JUSTIFICATIONS TO CUT OUT FOR PROOF PUZZLE

Transitive Property (,)

$$m\angle DHE + m\angle EHF = m\angle DHF$$

Given

Angle Addition Postulate

$$m\angle AHB = m\angle FHG$$

$$m\angle CHD + m\angle DHE + m\angle EHF = m\angle CHF$$

Angle Addition Postulate

Transitive Property (,)

$$m\angle BHC + m\angle CHD = m\angle CHD + m\angle EHF$$

Substitution (,)

$$m\angle BHD = m\angle DHF$$

Given

Subtraction Property of Equality

$$m\angle AHB = m\angle EHF$$

Substitution (,)

$$m\angle CHD + m\angle EHF = m\angle DHF$$

Given

$$m\angle BHC + m\angle CHD = m\angle BHD$$

Substitution (,)

$$m\angle CHF = m\angle BHE$$

$$m\angle CHD + m\angle EHF = m\angle BHD$$

Transitive Property (,)

Angle Addition Postulate

$$m\angle EHF + m\angle CHD + m\angle DHE = m\angle BHE$$

$$m\angle CHD = m\angle DHE$$

$$m\angle BHC = m\angle EHF$$

Substitution (,)

Substitution (,)

Angle Addition Postulate

$$m\angle BHC + m\angle CHD + m\angle DHE = m\angle BHE$$

Given

Addition Property of Equality



QUIZ: WRITING TWO-COLUMN PROOFS

Complete each proof.

Name: _____

Date: _____ Class: _____

1. Given: $a + c = 3f$

$$c = d$$

$$a = 2d$$

Prove: $d = f$

2. Given: $x = y$

$$z = y$$

$$w = x$$

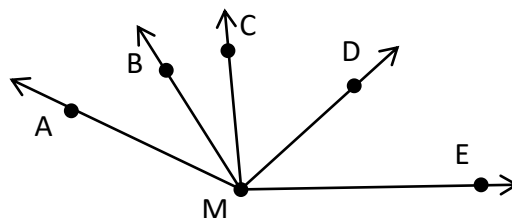
$$w = t$$

Prove: $t = z$

	STATEMENT	JUSTIFICATION		STATEMENT	JUSTIFICATION
1			1		
2			2		
3			3		
4			4		
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6			6		
7			7		
8			8		

3. Given: $\overrightarrow{MD}$ bisects $\angle CME$, $\angle AMB \cong \angle BMC$

Prove: $m\angle BMD = m\angle AMB + m\angle DME$

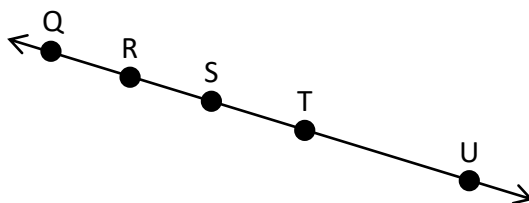


	STATEMENT	JUSTIFICATION
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4. Given: $QT = SU$

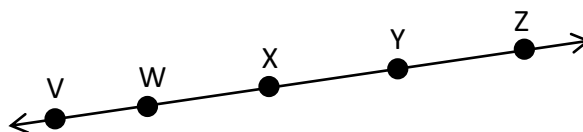
Prove: $QS = TU$



	STATEMENT	JUSTIFICATION
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5. Given: X is the midpoint of $\overline{VZ}$ and of $\overline{WY}$
 W is the midpoint of $\overline{VX}$
 Y is the midpoint of $\overline{XZ}$

Prove: $VY = WZ$



	STATEMENT	JUSTIFICATION
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13		

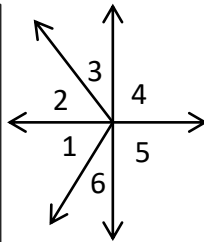


PROOFS USING SPECIAL ANGLE PAIRS

Name: _____
Date: _____ Class: _____

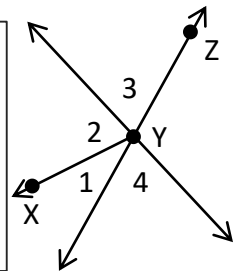
1. Given: $\angle 4$ is a right angle,
 $\angle 1 \cong \angle 2$

Prove: $\angle 2$ and $\angle 6$ are complementary.



2. Given: $\angle 2$ is a right angle,
 $m\angle 4 = 60^\circ$

Prove: $m\angle XYZ = 150^\circ$

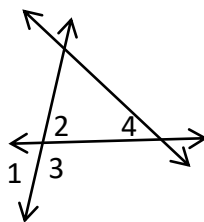


	STATEMENT	JUSTIFICATION
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	STATEMENT	JUSTIFICATION
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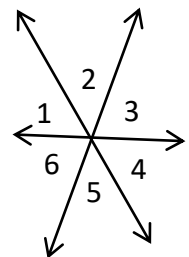
3. Given: $\angle 3$ and $\angle 4$ are supplements.

Prove: $\angle 2 \cong \angle 4$



4. Given: $\angle 1 \cong \angle 2$

Prove: $\angle 5 \cong \angle 4$



	STATEMENT	JUSTIFICATION
1		
2		
3		
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8		

	STATEMENT	JUSTIFICATION
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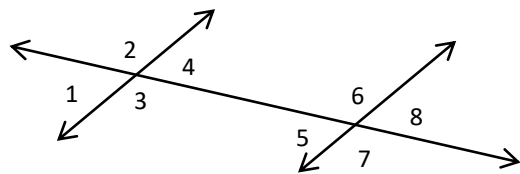


GEOMETRY PROOFS USING SPECIAL ANGLE PAIRS WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Given: $\angle 4 \cong \angle 8$

Prove: $\angle 4 \cong \angle 5$



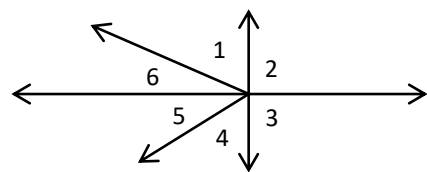
	STATEMENT	JUSTIFICATION
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GEOMETRY PROOFS USING SPECIAL ANGLE PAIRS WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

Given: $\angle 2$ is a right angle, $\angle 4 \cong \angle 6$

Prove: $\angle 5$ and $\angle 6$ are complementary.



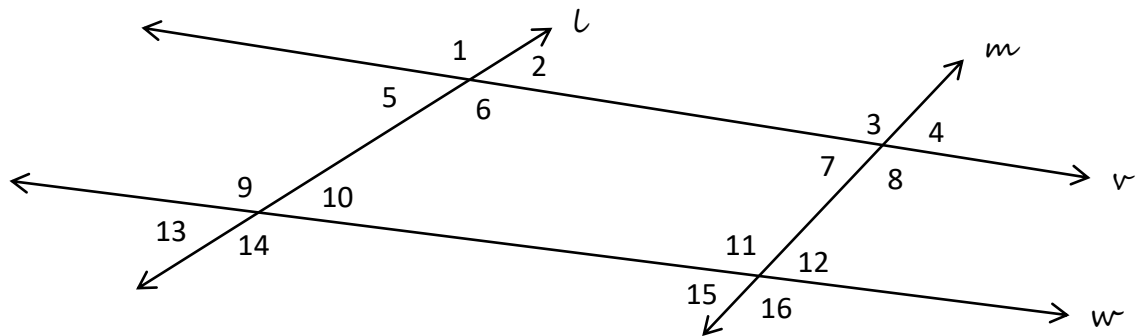
	STATEMENT	JUSTIFICATION
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PROOFS WITH SPECIAL ANGLES ALONG TRANSVERSALS

Name: _____

Date: _____ Class: _____



1. Given: $\angle 2 \cong \angle 7$

Prove: $\angle 14 \cong \angle 16$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

2. Given: $v \parallel w$, $\angle 7 \cong \angle 11$

Prove: $m\angle 16 = 90^\circ$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		
9		

3. Given: $l \parallel m$, $v \parallel w$

Prove: $\angle 1 \cong \angle 16$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		
7		
8		

4. Given: $\angle 13 \cong \angle 4$, $l \parallel m$

Prove: $v \parallel w$

	STATEMENT	JUSTIFICATION
1		
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8		



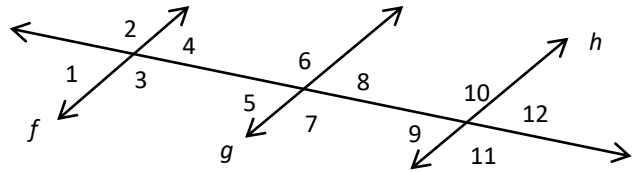
GEOMETRY PROOFS USING TRANSVERSALS

WARM-UP / EXIT TICKET A

Name: _____
Date: _____ Class: _____

Given: $\angle 11$ and $\angle 1$ are supplementary.

Prove: $\angle 3 \cong \angle 10$



	STATEMENT	JUSTIFICATION
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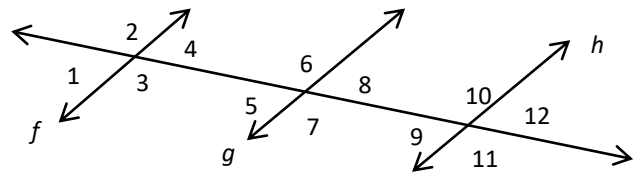
GEOMETRY PROOFS USING TRANSVERSALS

WARM-UP / EXIT TICKET B

Name: _____
Date: _____ Class: _____

Given: $\angle 12 \cong \angle 4$, $\angle 12 \cong \angle 5$

Prove: $m\angle 3 + m\angle 5 = 180^\circ$



	STATEMENT	JUSTIFICATION
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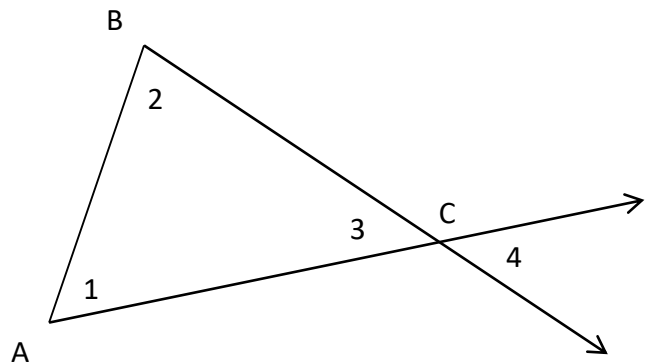
PROOFS WITH TRIANGLES

Name: _____
Date: _____ Class: _____

1. Given: $\angle 1 \cong \angle 4$

Prove: $\overline{AB} \cong \overline{BC}$

	STATEMENT	JUSTIFICATION
1		
2		
3		
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6		



2. Given: $m\angle 1 + m\angle 2 = 140^\circ$

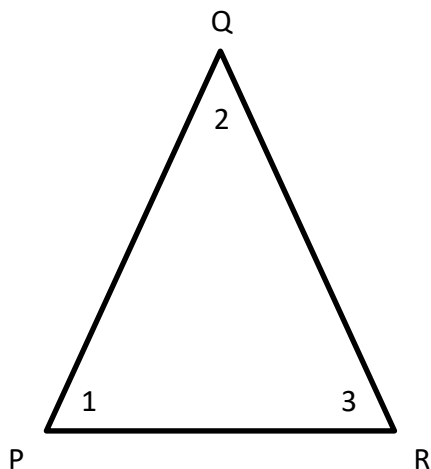
Prove: $m\angle 4 = 40^\circ$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		

3. Given: $AC = AB$

Prove: $m\angle 2 = m\angle 4$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		

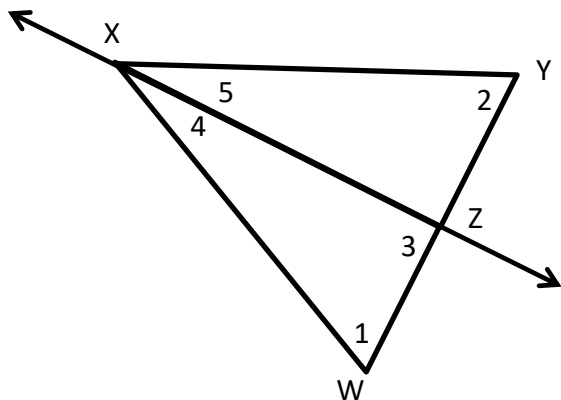


4. Given: $\angle 1 \cong \angle 3$, $\angle 1 \cong \angle 2$

Prove: $\overline{PQ} \cong \overline{PR}$

	STATEMENT	JUSTIFICATION
1		
2		
3		
4		
5		
6		





Name: _____
Date: _____ Class: _____

5. Given: $\angle 1 \cong \angle 2$, $XW = WY$

Prove: $WY = XY$

	STATEMENT	JUSTIFICATION
1		
2		
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6. Given: $WY = XW$

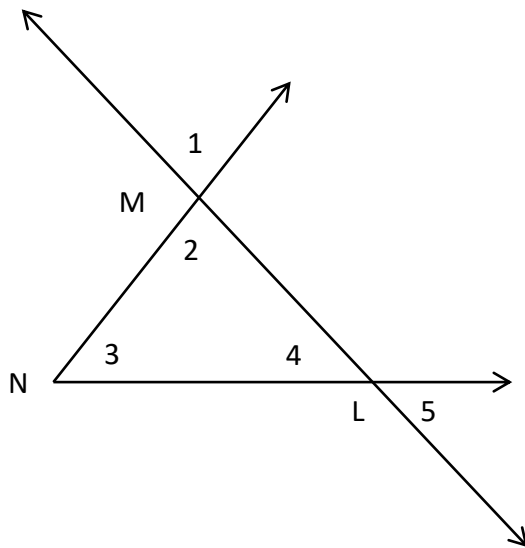
Prove: $m\angle 4 + m\angle 5 = m\angle 2$

	STATEMENT	JUSTIFICATION
1		
2		
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5		
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8		



QUIZ: PROOFS USING SPECIAL ANGLE
PAIRS AND TRIANGLES
Complete each proof.

Name: _____
Date: _____ Class: _____



1. Given: $\angle 1 \cong \angle 5$

Prove: $\overline{MN} \cong \overline{NL}$

	STATEMENT	JUSTIFICATION
1		
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2. Given: $\overline{MN} \cong \overline{ML}$

Prove: $\angle 3 \cong \angle 5$

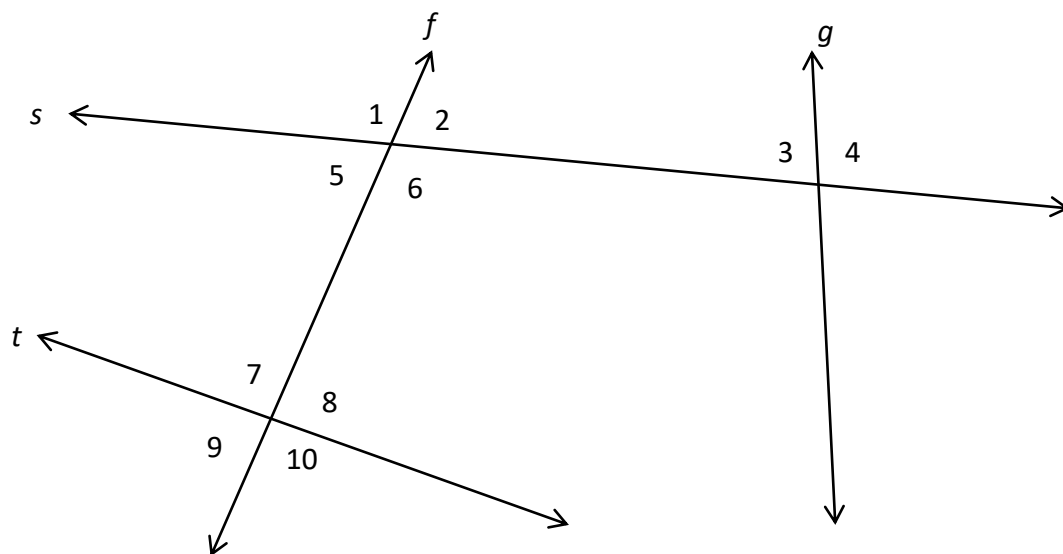
3. Given: $m\angle 2 = 90^\circ$, $m\angle 4 = 60^\circ$

Prove: $m\angle 3 = 30^\circ$

	STATEMENT	JUSTIFICATION
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	STATEMENT	JUSTIFICATION
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4. Given: $f \parallel g$, $\angle 3 \cong \angle 7$

Prove: $s \parallel t$

	STATEMENT	JUSTIFICATION
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5. Given: $\angle 8 \cong \angle 6$, $s \parallel t$

Prove: $m\angle 2 = 90^\circ$

	STATEMENT	JUSTIFICATION
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